YouTube Comment Toxicity Detector

Classification using Logistic Regression and DistilBERT.

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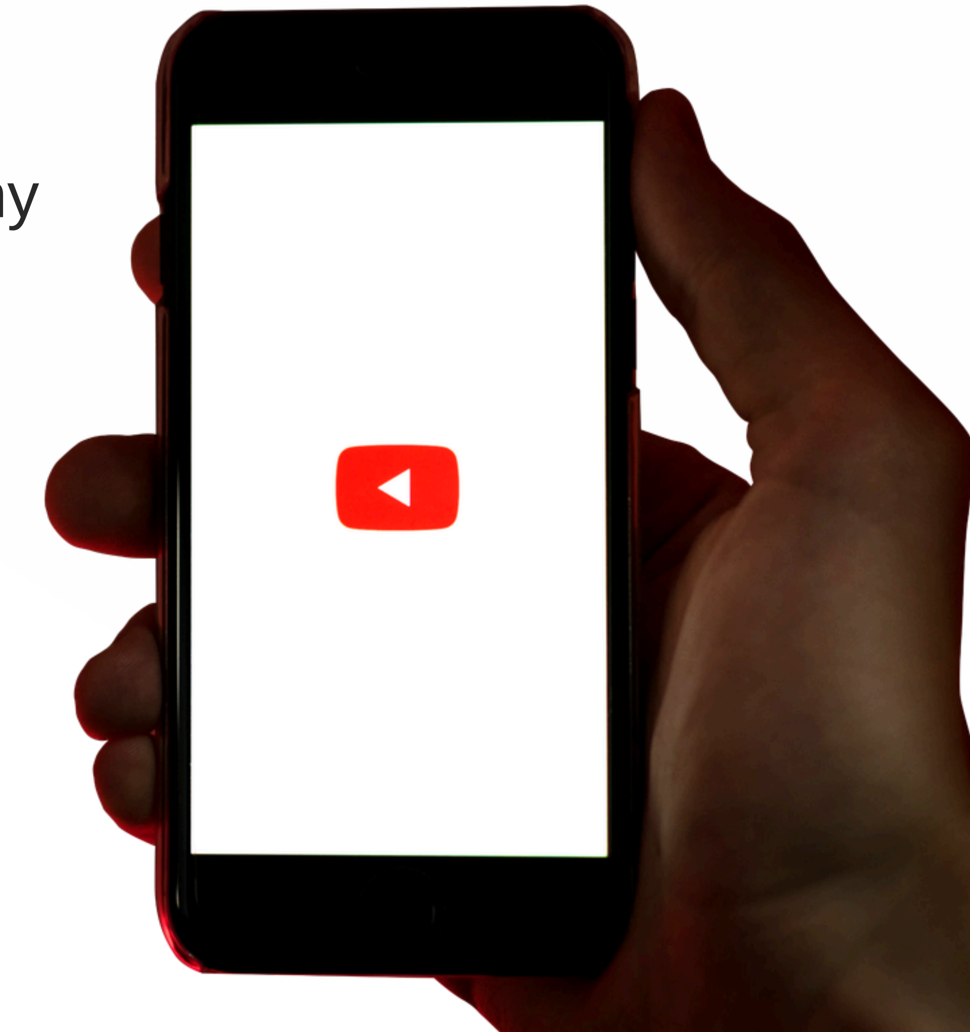
Problem Statement

1. **The Content Moderation Crisis:**
- Manual moderation of billions of daily YouTube comments is physically and economically impossible.
- Automated systems are essential to maintain safe digital environments and prevent the spread of hate speech.

2. **Extreme Class Imbalance:**
- The vast majority of comments in the dataset (over 143,000 out of 159,571) are non-toxic.
- Critical offensive categories, such as "Threat" and "Identity Hate," represent only a tiny fraction of the data, making it difficult for standard models to recognize them.

3. **Multi-Label Complexity:**
- Comments often belong to multiple categories simultaneously (e.g., "Obscene" and "Insult"), requiring a model that can predict six overlapping labels with high precision.



Dataset & The Multi-Label Challenge

- **Source**: Jigsaw Toxic Comment Classification Challenge dataset (provided by Google/Jigsaw).
- **Data Volume**: 159,571 training samples.
- **Target Labels**: The project addresses six non-exclusive categories: Toxic, Severe Toxic, Obscene, Threat, Insult, and Identity Hate.
- **Critical Problem:** Significant Class Imbalance. The vast majority of comments are "clean," while offensive categories like "Threat" and "Identity Hate" represent a tiny fraction of the data, necessitating specialized handling.

Dataset

Preprocessing & Vectorization Pipeline

Text Normalization

Utilized the SpaCy library for advanced Lemmatization and removal of non-essential components like stopwords and URLs.

Statistical Path

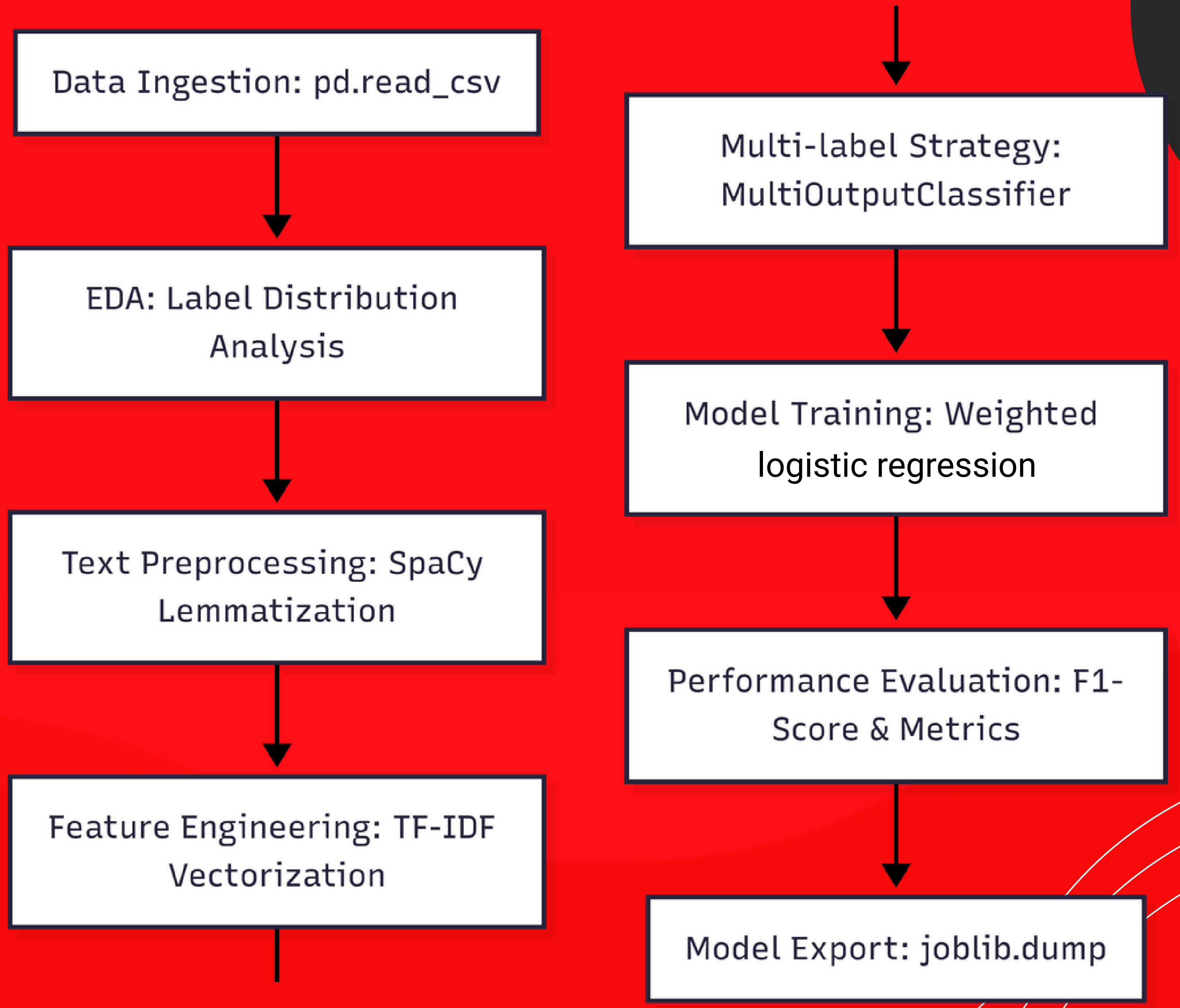
Implemented a TF-IDF Vectorizer limited to the top 10,000 features to convert text into numerical weights based on word frequency.

Neural Path

Employed the DistilBERT Tokenizer using subword tokenization with a maximum sequence length of 128 tokens to capture contextual meaning.

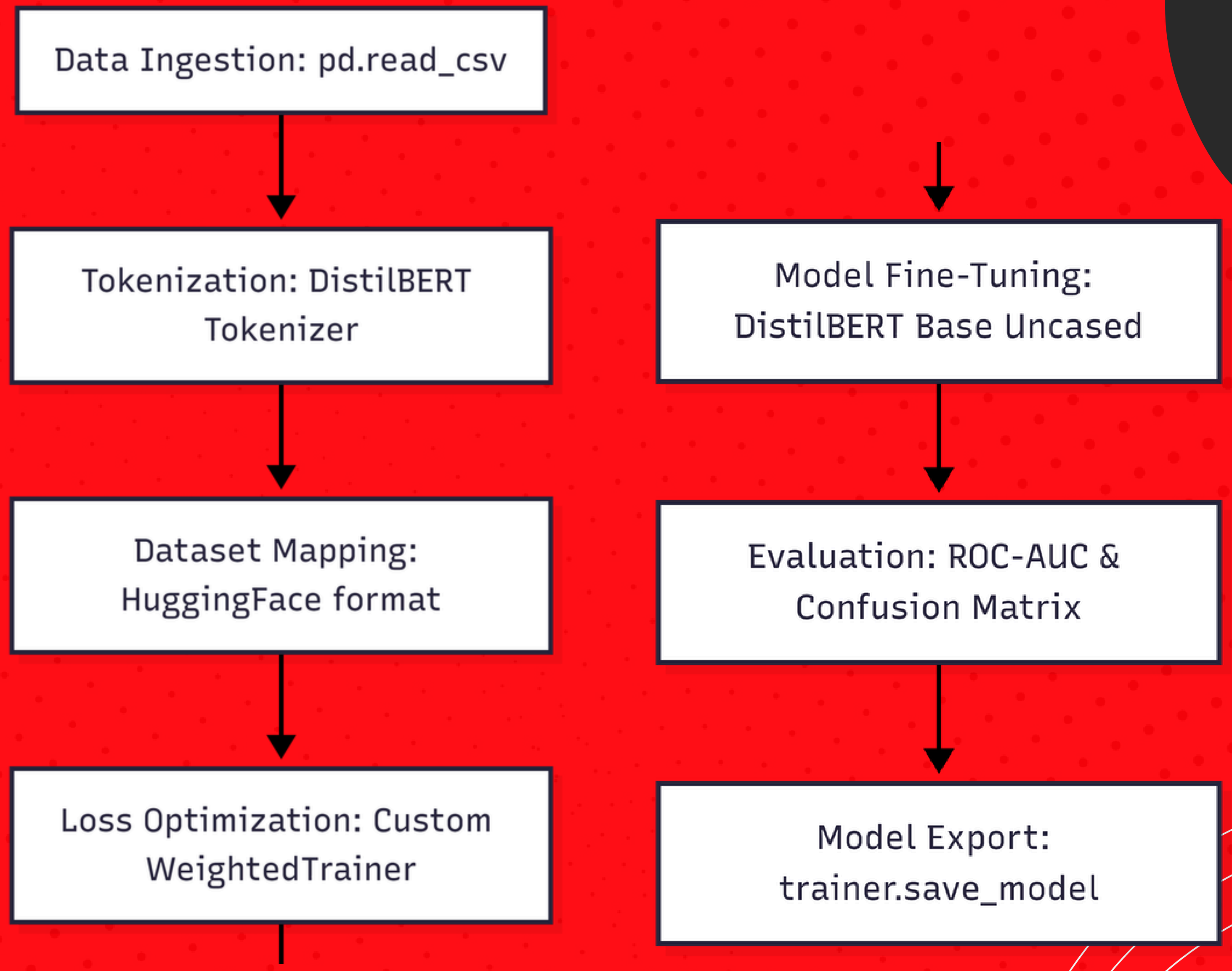
Baseline Architecture: Weighted Logistic Regression

- **Approach**: Utilized a logistic regression model wrapped in a MultiOutputClassifier to handle multiple target labels simultaneously.
- **Imbalance Mitigation**: Applied the class_weight='balanced' parameter to automatically adjust weights inversely proportional to class frequencies.
- **Rationale**: logistic regression was selected as the baseline because it is computationally efficient, fast for inference, and highly effective when paired with TF-IDF for large-scale text data.

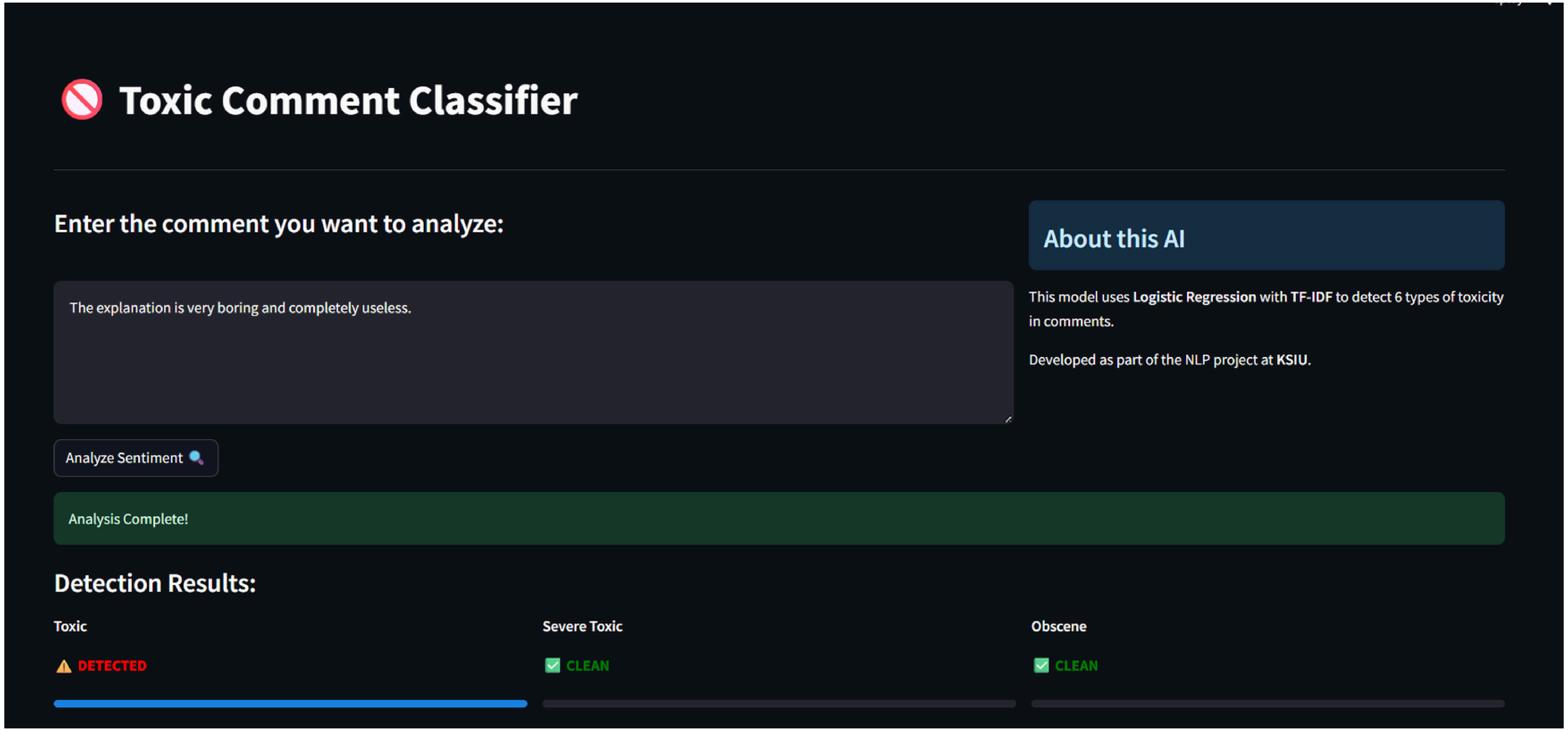


Advanced Architecture: DistilBERT Fine-tuning

- **Model Selection**: distilbert-base-uncased. It is 40% smaller and 60% faster than BERT while retaining 97% of its language understanding performance.
- **Custom Training Logic:** Developed a specialized WeightedTrainer that incorporates pos_weight within the BCEWithLogitsLoss function. This forces the model to prioritize learning from rare, high-severity labels.
- **Training Dynamics:** Used a learning rate of 2e-5 with a Cosine Learning Rate Scheduler and a 10% warmup phase to ensure training stability and prevent overfitting.



Model Deployment: Streamlit Dashboard (Baseline)



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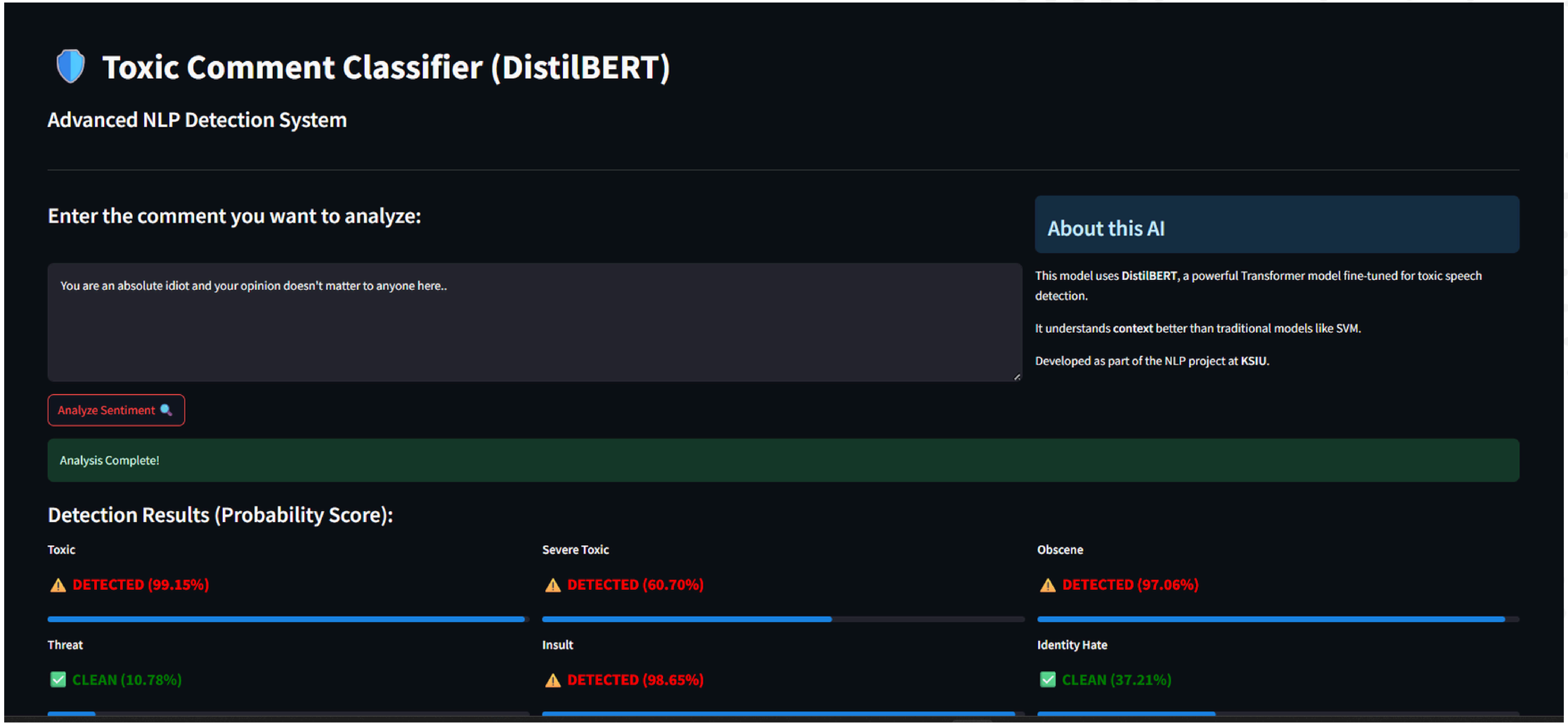

Conclusion & Future Scope

- Conclusion: The project successfully developed a robust detector where DistilBERT outperformed traditional models by capturing semantic intent rather than just word frequency.
- Deployment Potential: The model is ready for integration with the YouTube Data API and a Streamlit dashboard for real-time comment moderation.
- Future Directions: Further research will explore larger models like RoBERTa and the use of Data Augmentation techniques to generate more synthetic samples for the rarest labels.

Thank You